

Saple Project: Mayday Episodes Data Analysis

From Data Extraction to Predictive Modeling

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Project Pipeline

- ① **Scrape Data:** Extract episode details from Wikipedia.
- ② **Database Design:** plain CSV or (or SQL or NoSQL Solution)
- ③ **EDA & Visualization:** Trends in accidents/fatalities.
- ④ **Feature Engineering:** Prepare for modeling.
- ⑤ **Predictive Modeling:** Accident severity classification.
- ⑥ **Validation:** Test on external dataset (Aviation Safety Network).

EDA and Visualization

Uncover trends, anomalies, and relationships in Mayday episode data.

fatalities, date, aircraft type, error type??

No. overall	No. in series	Title	Incident	Original release date
1	1	"Unlocking Disaster"	United Airlines Flight 811	10 September 2003

On 24 February 1989, part of the right-side fuselage of United Airlines Flight 811 rips off, ejecting nine people from the aircraft and causing an explosive decompression. The flight later lands safely at Honolulu without any more loss of life. It was later determined that an electrical short circuit caused the cargo door to open.

Note: This accident is covered again in the Season 24 episode 'Terror Over The Pacific'.

Type of aircraft: Boeing 747-100

2	2	"Racing the Storm"	American Airlines Flight 1420	3 September 2003
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On 1 June 1999, American Airlines Flight 1420 tries to land at the Little Rock National Airport during a storm, but overruns the runway, killing 11 people. The crash was caused by the crew not arming the spoilers during their pre-landing checks.

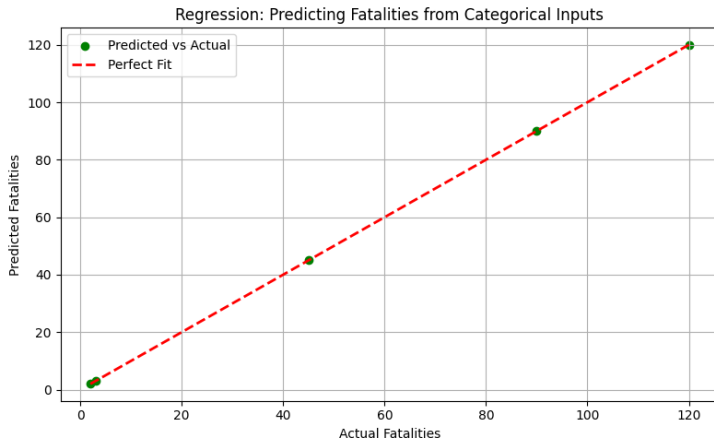
Note: Stephen Bogaert, the narrator for the Canadian version, also played First Officer Origel.

Type of aircraft: McDonnell Douglas MD-82

Key Analyses

- **Accident Causes:** Frequency of mechanical vs. human error.
- **Fatalities:** Distribution by season/year.
- **Episode Trends:** Air dates vs. accident severity.

Task 1: Predict Fatalities Using Regression



Inputs (X): accident_type, aircraft_model,...

Output (y): fatality_count (numeric)

Task 1: Predict Fatalities Using Regression

- **Structured Features:**

- Accident type (mechanical/human error)
- Aircraft model and age
- Weather conditions (if available)

- **Derived Features:**

- Seasonality (from air dates)
- Keyword frequencies from episode summaries

Model :

- Linear Regression (baseline)

Validation:

- RMSE and MAE as primary metrics
- Cross-validation on Mayday data
- Final test on Aviation Safety Network

Task 2: : Accident Severity Prediction

Target Variable: Accident Severity (e.g., Low/Medium/High).

Features:

- **Structured:** Fatalities count, accident type (mechanical/human error).
- **Text:** Episode summaries (TF-IDF, word embeddings).
- **Temporal:** Air date (seasonality effects).

Preprocessing:

- Handle missing data (e.g., impute fatalities).
- Normalize numerical features.

Task 2: : Accident Severity Prediction

Approach: Supervised learning (Classification).

Candidate Models:

- Logistic Regression (Baseline).
- Gradient Boosting (XGBoost).

Metrics:

- Accuracy, Precision/Recall (class imbalance).
- Confusion matrix.

Task 2: : Accident Severity Prediction

Validation: Generalization Test Dataset: Aviation Safety Network

(ASN)

<https://aviation-safety.net/database/>

acc. date	type	reg.	operator	fat.	location	dmg
2 Aug 1919	Caproni Ca.48		Caproni	14	Verona	 w/o
11 Aug 1919	Felixstowe Fury	N123	Royal Air Force - RAF	1	off Felixtowe RNAS	 w/o

Outcome:

- If performance holds, model generalizes.
- Else, refine features/training data.

1. Documentation:

- **Technical Report:** Detailed methodology, EDA insights, and model performance metrics.

2. Visual Summaries:

- **Interactive Dashboard:** Visualizations of accident trends.

3. Model Artifacts:

- **Trained Models:** Saved models for deployment.